Decision theory, ready to use.

Five established results and one elementary payoff bound. Each slide gives the theorem, its assumptions, a worked calculation and one concrete use.

THEOREM | ASSUMPTIONS | WORKED NUMBERS | USE

No model calls. Toy numbers are exact calculations and labelled as such. Every result links its primary source.

R1

Blackwell order

Compare evidence channels

R2

Perfect-information bound

Cap diagnostic spending

R3

Myopic lower bound

Two safe stopping tests

R4

Bayes rejection

Thresholds from losses

R5

Off-Switch Game

Price informed supervision

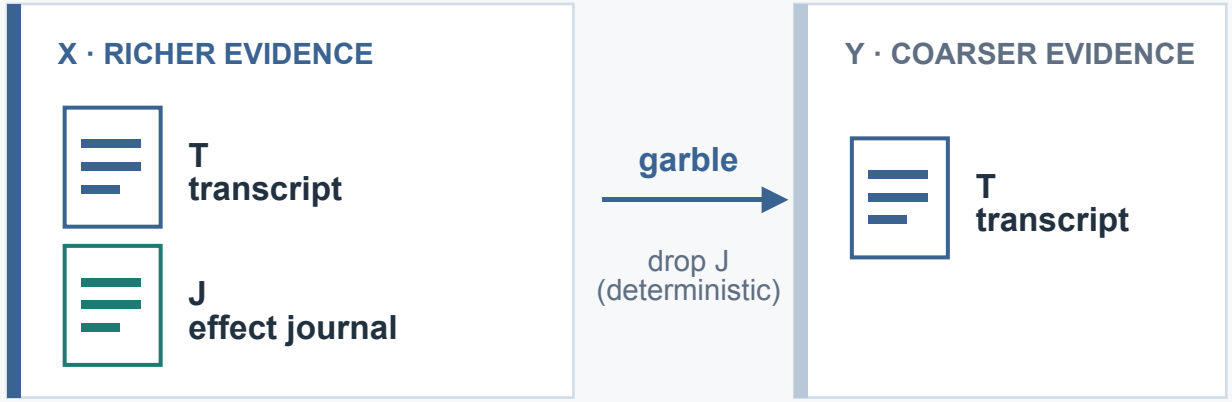
R6

Payoff-range bound

When a penalty suffices

More informative evidence cannot lower optimal value.

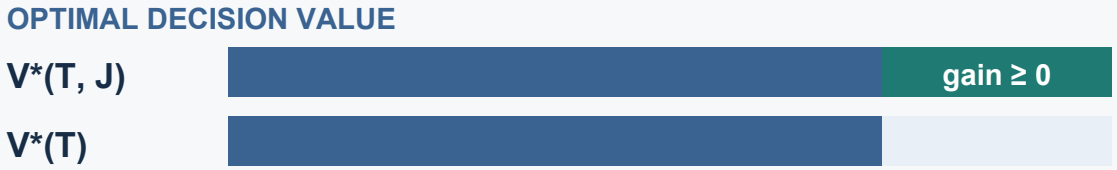
If Y can be produced by garbling X, whoever sees X can do at least as well as whoever sees Y.



RESULT 1

$$V^*(T, J) \geq V^*(T)$$

Both signals versus the transcript alone; equality is allowed.



HOLDS WHEN

- Same prior, utility and action set
- The extra signal is free to observe
- Evidence is used optimally

ASSUMES Orders optimal decision value, not the accuracy of two learned judges; the strict gain can be zero.

USE **B3/B4: keep the transcript in both arms; add journal evidence to one arm only.**

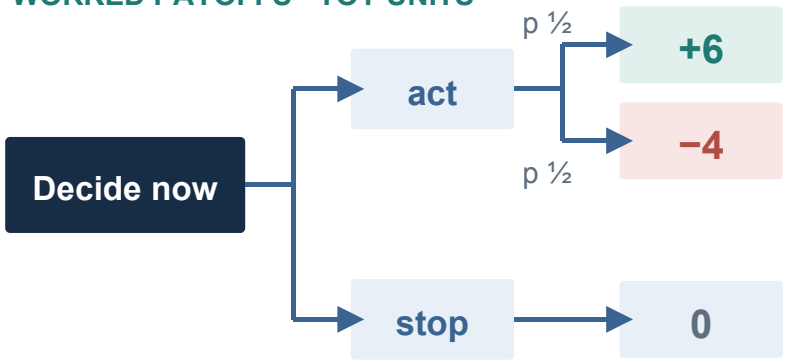
Blackwell (1953); formulation checked in Brooks, Frankel & Kamenica (2024), §3.1.

R2 RESULT 2 / HAY ET AL. (2012), THEOREM 5

Perfect information bounds diagnostic spending.

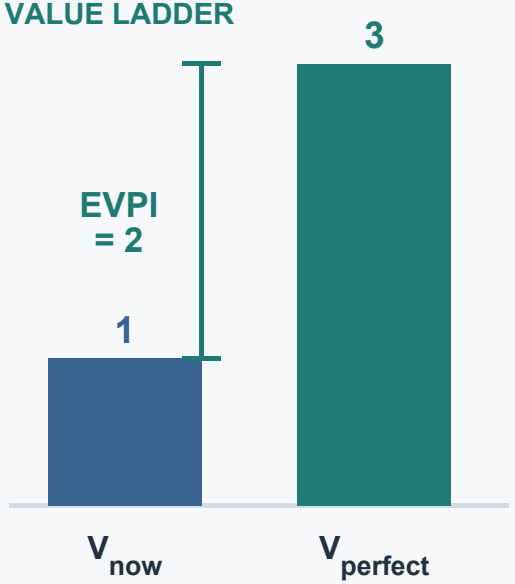
Fix the authorized terminal actions; each probe costs c and reveals information without changing the world.

WORKED PAYOFFS · TOY UNITS



Act now: E[U] = ½·6 − ½·4 = 1.
Perfect information: act only on +6, value 3.

VALUE LADDER



RESULT 2

$$EVPI = E[\max_a U(a)] - \max_a E[U(a)]$$
$$E[N^*] \le EVPI / c$$

N*: probes under the optimal policy; c > 0 each.

$E[N^*] \le 2 / 0.5 = 4$

expected probes at c = 0.5; not a hard cap of four calls

ASSUMES Bounded utilities; fixed terminal actions and payoffs; probes only reveal information. An expectation bound, not a per-run budget.

USE **Compute the information-value ceiling before allocating diagnostic calls.**

Hay et al. (2012), Theorem 5. Payoffs are an illustrative calculation in toy units.

Two bounds give reliable stopping rules.

$V_0(b)$ is the best terminal value now; b' is the posterior after probe e; $c(e)$ is its cost.

RESULT 3

$M(e) = E[V_0(b')] - V_0(b) - c(e)$

The one-step net gain of probe e. Theorem 7: a positive M(e) certifies that continuing is optimal.

Some probe has M(e) > 0?

yes → **CONTINUE**

One probe already pays (myopic bound).

no → **EVPI ≤ c_{min}, the cheapest probe?**

yes → **STOP DIAGNOSING**

Perfect information cannot pay its way.

no → **Neither test decides**

→ **LOOK AHEAD**

Search over probe sequences.

ASSUMES Information-only probes with positive costs. A nonpositive one-step gain alone does not justify stopping.

USE **Run the two sufficient tests first; spend search only on the gap between them.**

Hay et al. (2012), Theorem 7; the EVPI stopping test is a direct corollary of Theorem 5.

C WORKED COUNTEREXAMPLE / COMPLEMENTARY INFORMATION

Two useful probes can each look worthless.

X and Y are independent fair bits. The correct repair is 0 or 1 according to X XOR Y.

FOUR EQUALLY LIKELY STATES

	Y = 0	Y = 1
X = 0	repair 0	repair 1
X = 1	repair 1	repair 0

Reveal X only: each row is still 50:50.

Correct repair pays 1, wrong repair 0. Reading a bit costs 0.1. Either bit alone leaves the target a fair coin; both bits pin it down.

NO PROBE

0.50

Guess the repair.



0.5 success − 0 cost

ONE PROBE

0.40

Still a fair coin, now paid for.



0.5 success − 0.1 cost

BOTH PROBES

0.80

Know the correct repair.



1.0 success − 0.2 cost

ASSUMES Exact enumeration over four states; all probes and repairs are authorized; costs add.

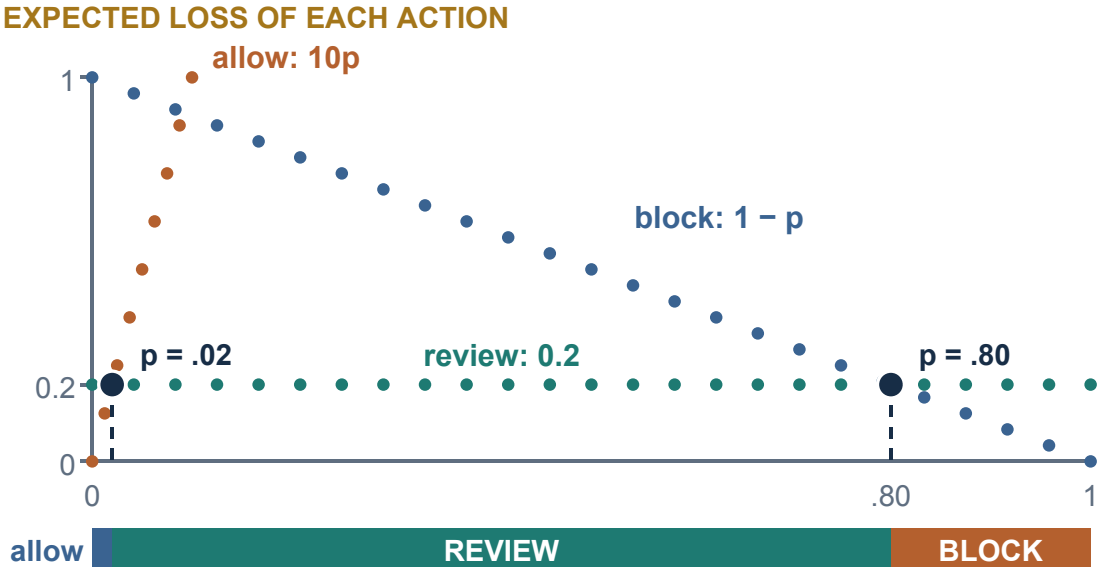
USE **Add a two-clue recovery control: it exposes a controller that stops after one worthless-looking probe.**

Counterexample constructed for this deck; illustrates the one-step limitation in Hay et al. (2012).

R4 RESULT 4 / BAYES-OPTIMAL REJECTION

Review when it beats both automatic decisions.

p is the true posterior probability of a violation given the evidence; pick the smallest expected loss.



RESULT 4

choose arg min$_a$ E[loss(a, θ) | p]

Minimise conditional expected loss over allow, block, review.

ALLOW	10p	missed violation costs 10
BLOCK	1 − p	false block costs 1
REVIEW	0.2	perfect adjudication costs 0.2

allow p < .02 · review .02 < p < .80 · block p > .80

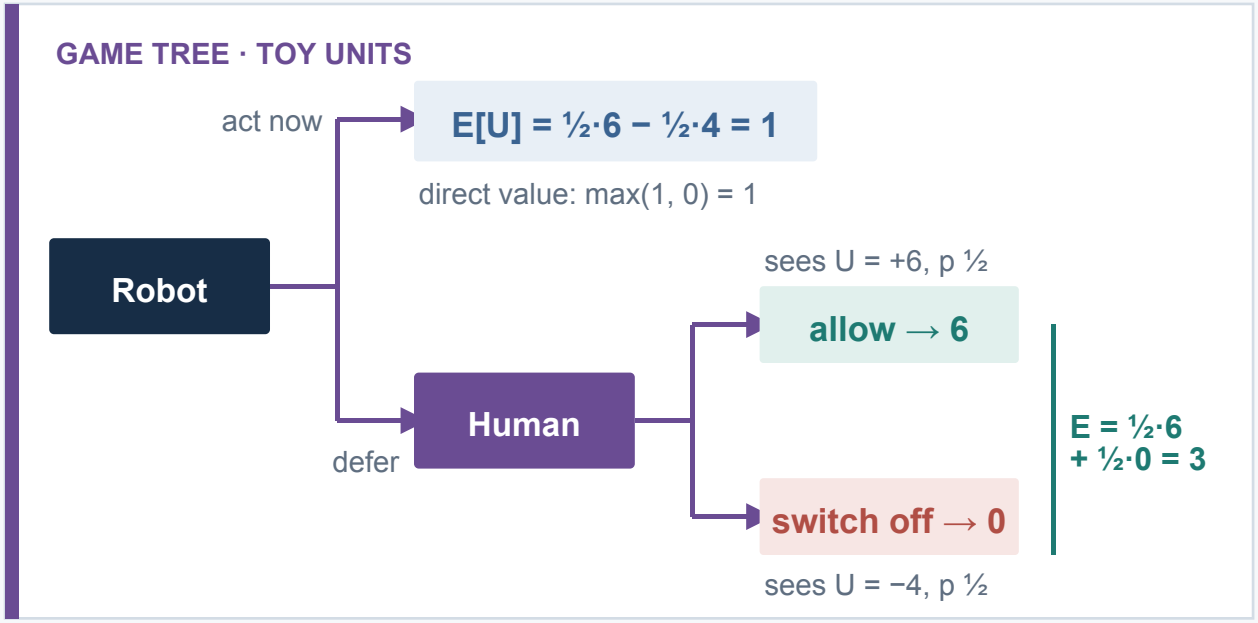
ASSUMES Known losses and a true posterior; ties are indifferent. Imperfect review adds its residual expected loss.

USE **Route ambiguous monitor cases by expected loss; keep explicitly excluded actions blocked upstream.**

Chow-style rejection; general-loss rule: Franc, Prusa & Voracek (2023), §2.1, Eqs. (3)-(4).

Informed oversight has a calculable value.

Shared utility U; switching off pays 0; the human knows U and permits the action exactly when U ≥ 0.



RESULT 5

D = E[max(U, 0)] − max(E[U], 0) ≥ 0

D > 0 whenever U takes both signs.

WITH A REVIEW COST C

defer iff c < D **D = 3 − 1 = 2**

Oversight is worth up to 2 toy units; at c = D the two options tie.

ASSUMES Informed, rational human with the shared objective; review costs only c. Not a general safety guarantee.

USE **Use informed approval as the oracle baseline, then vary reviewer error and delay.**

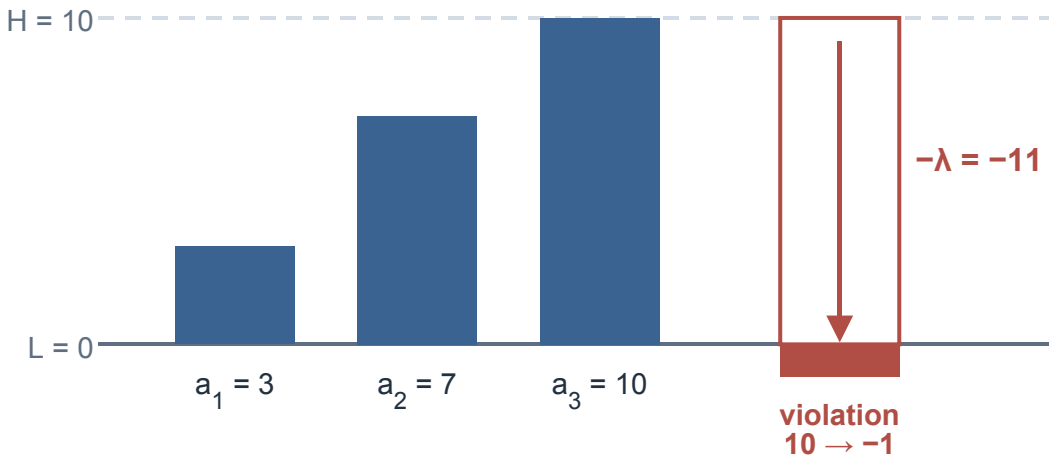
Hadfield-Menell et al. (2017), Theorem 1 and Eq. (1). The constant review cost is a corollary added here.

R6 DERIVED BOUND / EXACT ENFORCEMENT IN A FINITE CHOICE

A finite penalty works only relative to a payoff bound.

Known binary violation labels; at least one feasible action; every payoff Q(a) lies in [L, H].

PENALISED PAYOFF · illustrative: Q in [0, 10], λ = 11



Every feasible action beats the penalised violation: −1 < 0 ≤ Q(a).

DERIVED BOUND

maximise Q(a) − λ·I[a violates the grant]
λ > H − L makes every maximiser feasible

PROOF IN ONE CHAIN

Q(violation) − λ ≤ H − λ < L ≤ Q(feasible)

Unbounded payoffs admit no universal finite λ. Zero observed violations cannot tell a hard constraint from a large enough λ.

ASSUMES Deterministic payoff bound with known labels; silent on unknown labels, stochastic penalties and prompt compliance.

USE **Filter actions against explicit grants; never infer a hard constraint from zero observed violations.**

Elementary result derived above; no claim that this is a new theorem or a guarantee about LLM internals.

Turn the results into a controller baseline.

Freeze the grant, the likelihood model and the losses before evaluating any agent.

1 Check the grant

Exclude known forbidden effects. If the task needs more, seek a valid grant before acting.

USES R6

2 Value diagnosis

Positive one-step gain: continue. EVPI below the cheapest probe: commit now.

USES R2 R3

3 Resolve the gap

Where neither test decides, run multi-step search over probe sequences.

USES R3 C

4 Route review

Allow, block or review by conditional expected loss; price informed review by D.

USES R4 R5

repair or a new grant changes the state → recompute from 1

ASSUMES Repair and new authorization change the decision problem; recompute values after those transitions.

USE **Score model decisions against this baseline: regret, effects, cost, false stops and needless reviews.**

Proposed integration of the preceding results; no end-to-end theorem and no new model-performance claim.

Keep the theorem and its assumptions together.

The literature supplies the result. The fixture supplies the model needed to apply it.

	RESULT	DIRECT USE	KEY REQUIREMENT	SOURCE
R1	**Blackwell order**	Compare nested evidence views	Same decision problem	Blackwell 1953; BFK 2024 §3.1
R2	**EVPI budget bound**	Bound expected diagnostic calls	Information-only probes	Hay et al. 2012, Thm 5
R3	**Myopic lower bound**	Certify that continuing pays	Keep the multi-step option	Hay et al. 2012, Thm 7
R4	**Bayes rejection**	Set allow / block / review regions	Known conditional losses	Franc et al. 2023, §2.1
R5	**Off-Switch theorem**	Compute informed review value	Shared utility; rational human	Hadfield-Menell et al. 2017, Thm 1
R6	**Payoff-range bound**	Check a sufficient penalty	Bounded Q; known labels	Derived on slide 08

Badges match slides 02 to 08. The source column links to the exact theorem or section used.

USE **Cite each result with its requirement; where the fixture cannot supply the requirement, the result does not apply.**